DR. IGOR KHOKHLOV

Assistant Professor of Cybersecurity

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~393	11	35+	5
CITATIONS	H-INDEX	PUBLICATIONS	APPS (GOOGLE PLAY)

Cybersecurity | Threat Modeling | AI / ML / Deep Learning | Android / Mobile Security | Data Quality & Governance | Embedded Systems / JNI

Distinguished cybersecurity scholar and computer scientist specializing in the fusion of artificial intelligence and information management. Research focus encompasses cyber-physical resilience, data quality metrics for IoT ecosystems, and the mitigation of sophisticated mobile application collusion. Proven track record in securing high-stakes infrastructure and engineering intelligent defensive frameworks for a modern digital landscape.

EXPERIENCE

2021 – Present — ACADEMIC

Assistant Professor of Cybersecurity

Sacred Heart University, School of Computer Science & Engineering

- Developing and teaching graduate and undergraduate cybersecurity courses.
- Conducting research in data quality, mobile security, and AI/ML applications.
- Supervising senior capstone projects and mentoring students.

2020 – 2021 — RESEARCH

Postdoctoral Research Scientist

Rochester Institute of Technology (RIT), Software Engineering Dept.

Led critical components of the DHS-funded 'Novel Vulnerability Intelligence Platform' (NVIP).

- Developed intelligent reconciliation algorithms for disparate CVE data streams.
- Researched adversarial threats to ML models, mapping mitigation strategies for mission-critical AI.

2016 – 2020 — RESEARCH

PhD Researcher / Research Assistant

Rochester Institute of Technology (RIT), Computer Science Dept.

Executed NSF-funded research focused on data quality evaluation in mobile ecosystems.

- Pioneered the 'D.I.F.E.N.S.E.' framework for automated Android security evaluation.
- Identified and modeled novel application collusion attack vectors in Android OS.

Summer 2018 — ACADEMIC

Adjunct Faculty

Rochester Institute of Technology, Computer Science Dept.

Independent instructor for graduate-level course "Foundations of Intelligent Systems" (MS in Computer Science program).

2013 – 2015 INDUSTRY

Embedded Team Leader

Dynamo Development

Oversaw engineering teams for complex Android/Windows embedded software solutions.

- Expert consultant on JNI/JNA technologies for low-level system integration.
- Developed secure communication protocols for Z-Wave wireless sensor nodes.

2008 – 2013 INDUSTRY

Application Developer (Android, Windows)

EDAPS, JSC "KP OTI"

Developed secure mobile and desktop applications for identity verification systems.

- Android application for reading driver license data via NFC.
- Android application for reading vehicle technical passports with NFC.
- Android client and Windows server converting Android devices with NFC to contactless smart-card readers.
- Windows library for vehicle's technical state inspection card personalization.

EDUCATION

2016 – 2020 PHD

Ph.D. in Computing and Information Sciences

Rochester Institute of Technology, CCIS

Thesis: Integrated Framework for Data Quality and Security Evaluation on Mobile Devices.
GPA: 3.85 | Advisor: Dr. Leon Reznik

2005 – 2007 MASTERS

MS in Electronics (Honors)

National Technical University of Ukraine "KPI"

Focused on microsatellite space orientation and data processing.

2001 – 2005 BACHELORS

BS in Electronics (Honors)

National Technical University of Ukraine "KPI"

Foundation in electronic systems and engineering principles.

Languages

English: Native/Professional | **Ukrainian:** Native | **Russian:** Native

Toolkit

Java, Kotlin, Python, C++, SQL, Android Studio, PyTorch, LaTeX, Git

PUBLICATIONS

[1] 2025 **Weisman, O., & Khokhlov, I.** "Whisper through the Noise: A Privacy-Focused Messenger for Anonymous and Secure Communication". *Research Kernel / Preprint*.

[2] 2024 **Chuprov, S., Zatsarenko, R., Reznik, L., & Khokhlov, I.** "Data Quality Based Intelligent Instrument Selection with Security Integration". *ACM Journal of Data and Information Quality*, 16(3), 1-24.

[3] 2023 **Okutan, A., Mell, P., Mirakhorli, M., Khokhlov, I., Santos, J.C.S., Gonzalez, D., & Simmons, S.** "Empirical validation of automated vulnerability curation and characterization". *IEEE Transactions on Software Engineering*, 49(5), 3241-3260.

[4] 2022 **Khokhlov, I., Chuprov, S., & Reznik, L.** "Integrating Security with Accuracy Evaluation in Sensors Fusion". *IEEE Sensors*, pp. 1-4.

[5] 2022 **Chuprov, S., Reznik, L., Khokhlov, I., & Manghi, K.** "Multi-Modal Sensor Selection with Genetic Algorithms". *IEEE Sensors*, pp. 1-4.

[6] 2022 **Khokhlov, I., Okutan, A., Bryla, R., Simmons, S., & Mirakhorli, M.** "Automated Extraction of Software Names from Vulnerability Reports using LSTM and Expert System". *IEEE 29th Annual Software Technology Conference (STC)*, pp. 125-134.

[7] 2022 **Chuprov, S., Khokhlov, I., Reznik, L., & Shetty, S.** "Influence of transfer learning on machine learning systems robustness to data quality degradation". *International Joint Conference on Neural Networks (IJCNN)*, pp. 1-8.

[8] 2022 **Fazelnia, M., Khokhlov, I., & Mirakhorli, M.** "Attacks, defenses, and tools: A framework to facilitate robust AI/ML systems". *arXiv preprint arXiv:2202.09465*.

[9] 2021 **Chuprov, S., Viksnin, I., Kim, I., Melnikov, T., Reznik, L., & Khokhlov, I.** "Improving knowledge based detection of soft attacks against autonomous vehicles with reputation, trust and data quality service models". *IEEE International Conference on Smart Data Services (SMDS)*, pp. 115-120.

[10] 2020 **Khokhlov, I., Reznik, L., & Chuprov, S.** "Framework for Integral Data Quality and Security Evaluation in Smartphones". *IEEE Systems Journal*, 15(2), 2058-2065.

[11] 2020 **Khokhlov, I., Reznik, L., & Ajmera, S.** "Sensors in Mobile Devices Knowledge Base". *IEEE Sensors Letters*, 4(3), Art no. 5500404.

[12] 2020 **Alrubaye, H., Mkaouer, M.W., Khokhlov, I., Reznik, L., Ouni, A., & Mcgoff, J.** "Learning to Recommend Third-Party Library Migration Opportunities at the API Level". *Applied Soft Computing*, 90, 106140.

[13] 2020 **Khokhlov, I., Ligade, N., & Reznik, L.** "Recurrent Neural Networks for Colluded Applications Attack Detection in Android OS Devices". *International Joint Conference on Neural Networks (IJCNN)*, pp. 1-8.

[14] 2020 **Khokhlov, I., Reznik, L., & Lyshevski, S.** "Adaptive Data Fusion in Inertial Sensors and Data Quality of Sensor Networks". *IEEE 40th International Conference on Electronics and Nanotechnology (ELNANO)*.

[15] 2020 **Khokhlov, I. & Reznik, L.** "Knowledge Graph in Data Quality Evaluation for IoT Applications". *IEEE 6th World Forum on Internet of Things (WF-IoT)*, pp. 1-6. **(Outstanding Student Paper Award)**.

[16] 2020 **Chuprov, S., Marinenkov, E., Viksnin, I., Reznik, L., & Khokhlov, I.** "Image Processing in Autonomous Vehicle Model Positioning and Movement Control". *IEEE 6th World Forum on Internet of Things (WF-IoT)*, pp. 1-6.

[17] 2020 **Khokhlov, I. & Reznik, L.** "What is the Value of Data Value in Practical Security Applications". *IEEE Systems Security Symposium (SSS)*, pp. 1-8.

[18] 2020 **Chuprov, S., Viksnin, I., Kim, I., Reznik, L., & Khokhlov, I.** "Reputation and Trust Models with Data Quality Metrics for Improving Autonomous Vehicles Traffic Security and Safety". *IEEE Systems Security Symposium (SSS)*, pp. 1-8.

[19] 2019 **Khokhlov, I., Perez, M., & Reznik, L.** "Machine Learning in Anomaly Detection: Example of Colluded Applications Attack in Android Devices". *IEEE 18th International Conference On Machine Learning And Applications (ICMLA)*, pp. 1328-1333.

[20] 2019 **Khokhlov, I., Reznik, L., & Bhaskar, R.** "The Machine Learning Models For Activity Recognition Applications With Wearable Sensors". *IEEE 18th International Conference On Machine Learning And Applications (ICMLA)*, pp. 387-391.

[21] 2019 **Khokhlov, I., Perez, M., & Reznik, L.** "System Signals Monitoring And Processing For Colluded Application Attacks Detection In Android OS". *IEEE Western New York Image and Signal Processing Workshop (WNYISPW)*, pp. 1-5.

[22] 2019 **Khokhlov, I., Pudage, A., & Reznik, L.** "Sensor Selection Optimization with Genetic Algorithms". *IEEE SENSORS*, pp. 1-4.

[23] 2019 **Khokhlov, I., Li, Q., & Reznik, L.** "D.I.F.E.N.S.E.: Distributed Intelligent Framework for Expendable Android Security Evaluation". *14th Annual Symposium on Information Assurance (ASIA '19)*, pp. 18-27.

[24] 2018 **Khokhlov, I., Jain, C., Miller-Jacobson, B., Heyman, A., Reznik, L., & St. Jacques, R.** "MeetCI: A Computational Intelligence Software Design Automation Framework". *IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pp. 1-8.

[25] 2018 **Killawala, A., Khokhlov, I., & Reznik, L.** "Computational Intelligence Framework for Automatic Quiz Question Generation". *IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, pp. 1-8.

[26] 2018 **Khokhlov, I., Reznik, L., Cappos, J., & Bhaskar, R.** "Design of activity recognition systems with wearable sensors". *IEEE Sensors Applications Symposium (SAS)*, pp. 1-6.

[27] 2018 **Vora, A., Reznik, L., & Khokhlov, I.** "Mobile road pothole classification and reporting with data quality estimates". *Fourth International Conference on Mobile and Secure Services (MobiSecServ)*, pp. 1-6.

[28] 2018 **Khokhlov, I. & Reznik, L.** "Android System Security Evaluation". *IEEE 15th Annual Consumer Communications & Networking Conference (CCNC)*, pp. 1-2.

[29] 2018 **Khokhlov, I., Reznik, L., & Jothilingam, S.B.** "What Can Data Analysis Recommend on Design of Wearable Sensors?". *IEEE 15th Annual Consumer Communications & Networking Conference (CCNC)*, pp. 1-2.

[30] 2017 **Khokhlov, I. & Reznik, L.** "Colluded Applications Vulnerabilities in Android Devices". *IEEE 15th Intl Conf on Dependable, Autonomic and Secure Computing (DASC/PiCom/DataCom/CyberSciTech)*, pp. 462-469.

[31] 2017 **Khokhlov, I., Reznik, L., Kumar, A., Mookherjee, A., & Dalvi, R.** "Data Security and Quality Evaluation Framework: Implementation Empirical Study on Android Devices". *20th Conference of Open Innovations Association (FRUCT)*, pp. 161-168.

[32] 2017 **Khokhlov, I. & Reznik, L.** "Data Security Evaluation for Mobile Android Devices". *20th Conference of Open Innovations Association (FRUCT)*, pp. 154-160.

AWARDS & HONORS

- **Outstanding Student Paper Award** (2020) – IEEE World Forum on Internet of Things.
- **IEEE Student Complimentary Registration Grant** (2020) – IEEE World Congress on Computational Intelligence (WCCI).
- **NSF PhD Scholarship** (2016 – 2020) – National Science Foundation.
- **Honors Degrees** (BS & MS) – National Technical University of Ukraine.

RESEARCH GRANTS

- **NSF:** "Increasing Perseverance and Retention of Computing and Engineering Students Through Service" (Co-PI, 2022–2027).
- **CRDF Global:** "Security Evaluation and Improvement of the Personal Cyberinfrastructure with New Tools and Education Development" (Co-PI, 2021–2022).
- **US Military Academy/DoD:** "Self-learning capabilities for a mission-oriented data quality and security assurance in military IoT systems" (Co-PI, 2020–2021).

PROFESSIONAL SERVICE

Peer Reviewer: *IEEE Transactions on Dependable and Secure Computing*, *ACM Journal of Data and Information Quality*, *Elsevier Pervasive and Mobile Computing*, *PLOS ONE Journal*.

Community Engagement: Guest Speaker at Fairchild-Nichols Memorial Library (2023), Hamden Public Library (2021, 2022); Volunteer Mentor at ASA DataFest, RIT (2018).